LAPORAN PRAKTIKUM MOBILE PROGRAMMING

Pengaturan Layout Row dan Column di Flutter

Nama : Muhamad Kevin Dwi Arisandi

NIM : 240605110252

Kelas : Mobile Programming Praktikum A

Tanggal : 10 – 09 – 2026

JURUSAN TEKNIK INFORMATIKA

FAKULTAS SAINS DAN TEKNOLOGI

UNIVERSITAS ISLAM NEGERI MAULANA MALIK IBRAHIM

MALANG

GANJIL 2025/2026

I. Tujuan

- Memahami konsep main axis dan cross axis pada widget Row dan Column.
- Mengetahui penggunaan MainAxisSize.max dan MainAxisSize.min.
- Mempelajari nilai properti MainAxisAlignment untuk mengatur distribusi ruang.
- Mempelajari nilai properti CrossAxisAlignment untuk mengatur posisi pada sumbu silang.
- Mengimplementasikan properti-properti tersebut dalam kode Flutter.

II. Langkah Kerja

A. Persiapan Project

1. Membuat project Flutter baru dengan nama alignment.
2. Mengatur judul aplikasi sesuai dengan praktikum.
3. Menyiapkan struktur dasar aplikasi menggunakan MaterialApp, Scaffold, dan AppBar.

B. Pembuatan Custom Widget

1. Membuat widget KotakBiruJempol menggunakan Container.
2. Mengatur ukuran, warna, border, border radius, dan icon pada widget.
3. Membuat widget KotakBiruJempolKecil dengan ukuran yang lebih kecil.

C. Percobaan MainAxisSize

1. Menerapkan MainAxisSize.max pada Row dan Column.
2. Mengubah menjadi MainAxisSize.min.
3. Mengamati perubahan penggunaan ruang pada masing-masing layout.

D. Percobaan MainAxisAlignment

1. Menerapkan MainAxisAlignment.center.
2. Menguji start dan end.
3. Menguji spaceBetween, spaceAround, dan spaceEvenly.
4. Mengamati perubahan posisi widget pada Row dan Column.

E. Percobaan CrossAxisAlignment

1. Menerapkan CrossAxisAlignment.center.
2. Menguji start, end, dan stretch.
3. Menerapkan CrossAxisAlignment.baseline sesuai kebutuhan percobaan.
4. Mengamati perubahan posisi widget pada sumbu silang.

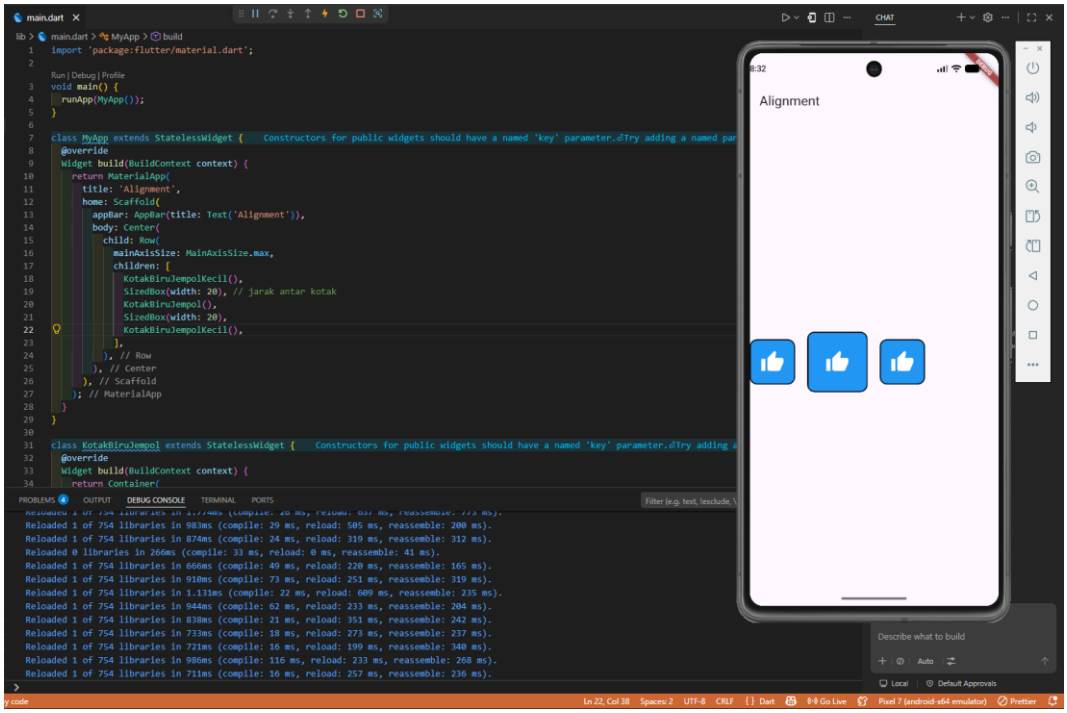
F. Pembuatan Tampilan Tugas

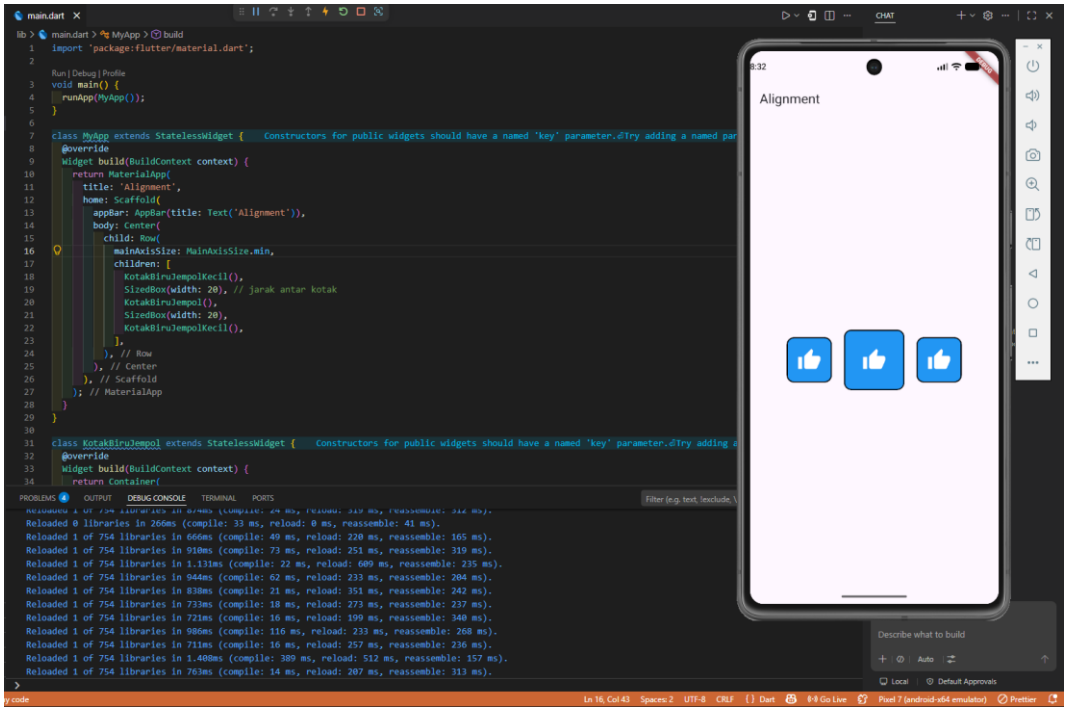
1. Membuat tampilan informasi cuaca menggunakan Column.
2. Menambahkan nama kota dan informasi suhu.
3. Menambahkan informasi prakiraan cuaca menggunakan Row.
4. Menggunakan icon sunny, cloudy_snowing, dan cloud.
5. Mengatur posisi icon dengan MainAxisAlignment.spaceEvenly.

III. Screenshot Hasil

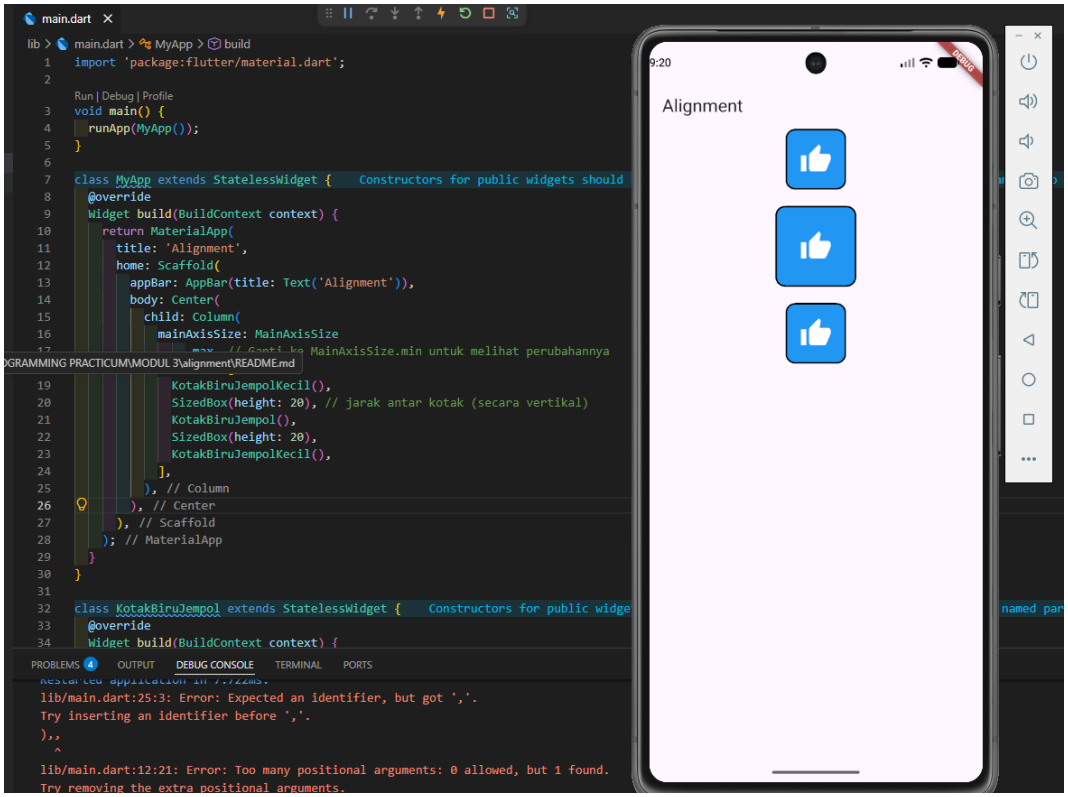
A. Percobaan MainAxisSize

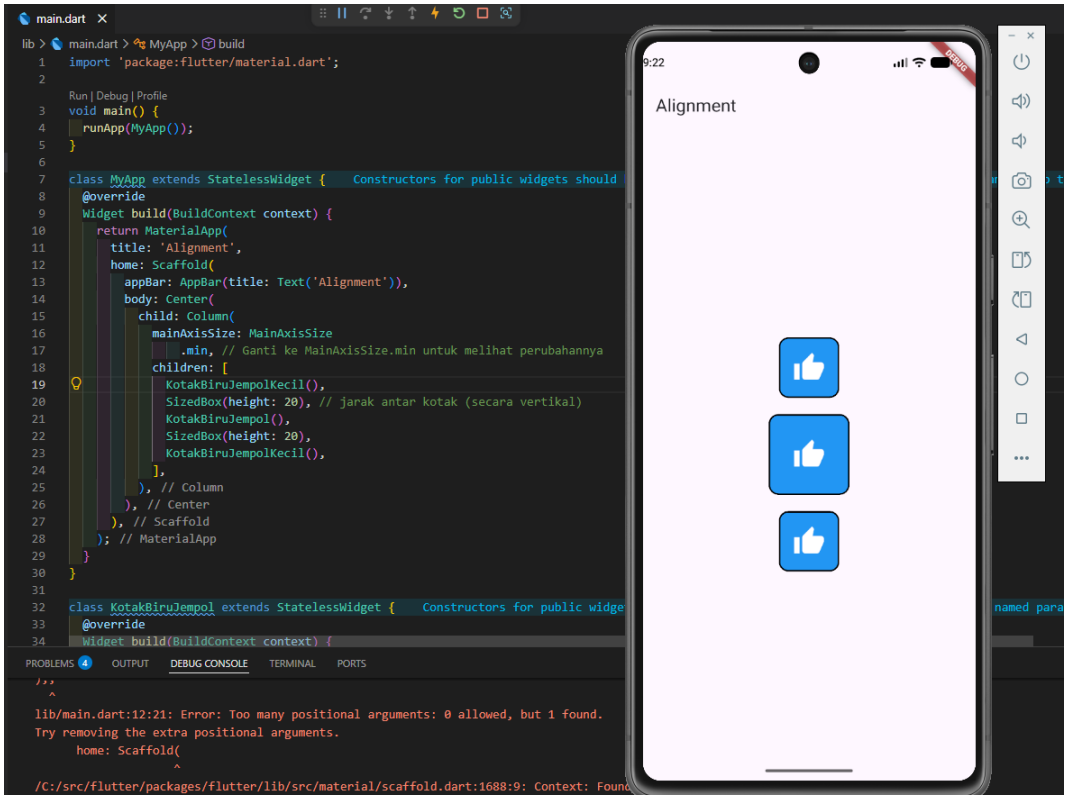
- Row




main.dart
lib > main.dart > MyApp > build
import 'package:flutter/material.dart';
Run | Debug | Profile
void main() {
runApp(MyApp());
}
class MyApp extends StatelessWidget {
@override
Widget build(BuildContext context) {
return MaterialApp(
title: 'Alignment',
home: Scaffold(
appBar: AppBar(title: Text('Alignment')),
body: Center(
child: Row(
mainAxisSize: MainAxisSize.min,
children: [
KotakBiruJempolKecil(),
SizedBox(width: 20), // jarak antar kotak
KotakBiruJempol(),
SizedBox(width: 20),
KotakBiruJempolKecil(),
],
), // Row
), // Center
), // Scaffold
); // MaterialApp
}
}
class KotakBiruJempol extends StatelessWidget {
@override
Widget build(BuildContext context) {
return Container(
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Reloaded 0 libraries in 266ms (compile: 33 ms, reload: 0 ms, reassemble: 41 ms).
Reloaded 1 of 754 libraries in 666ms (compile: 49 ms, reload: 220 ms, reassemble: 165 ms).
Reloaded 1 of 754 libraries in 910ms (compile: 73 ms, reload: 251 ms, reassemble: 319 ms).
Reloaded 1 of 754 libraries in 1.131ms (compile: 22 ms, reload: 609 ms, reassemble: 235 ms).
Reloaded 1 of 754 libraries in 944ms (compile: 62 ms, reload: 233 ms, reassemble: 204 ms).
Reloaded 1 of 754 libraries in 838ms (compile: 21 ms, reload: 351 ms, reassemble: 242 ms).
Reloaded 1 of 754 libraries in 733ms (compile: 18 ms, reload: 273 ms, reassemble: 237 ms).
Reloaded 1 of 754 libraries in 721ms (compile: 16 ms, reload: 199 ms, reassemble: 340 ms).
Reloaded 1 of 754 libraries in 986ms (compile: 116 ms, reload: 233 ms, reassemble: 268 ms).
Reloaded 1 of 754 libraries in 711ms (compile: 16 ms, reload: 257 ms, reassemble: 236 ms).
Reloaded 1 of 754 libraries in 1.408ms (compile: 389 ms, reload: 512 ms, reassemble: 157 ms).
Reloaded 1 of 754 libraries in 763ms (compile: 14 ms, reload: 207 ms, reassemble: 313 ms).
Alignment
Describe what to build

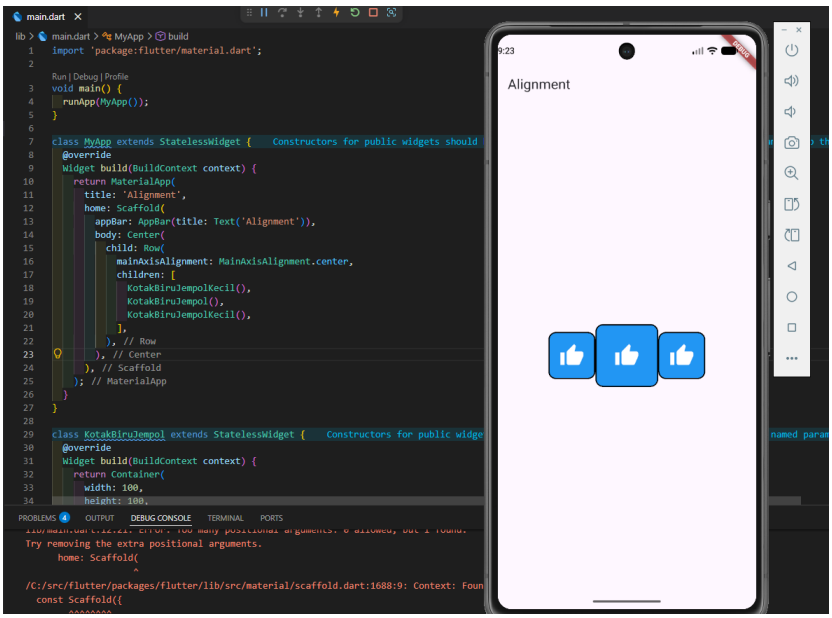
- Column

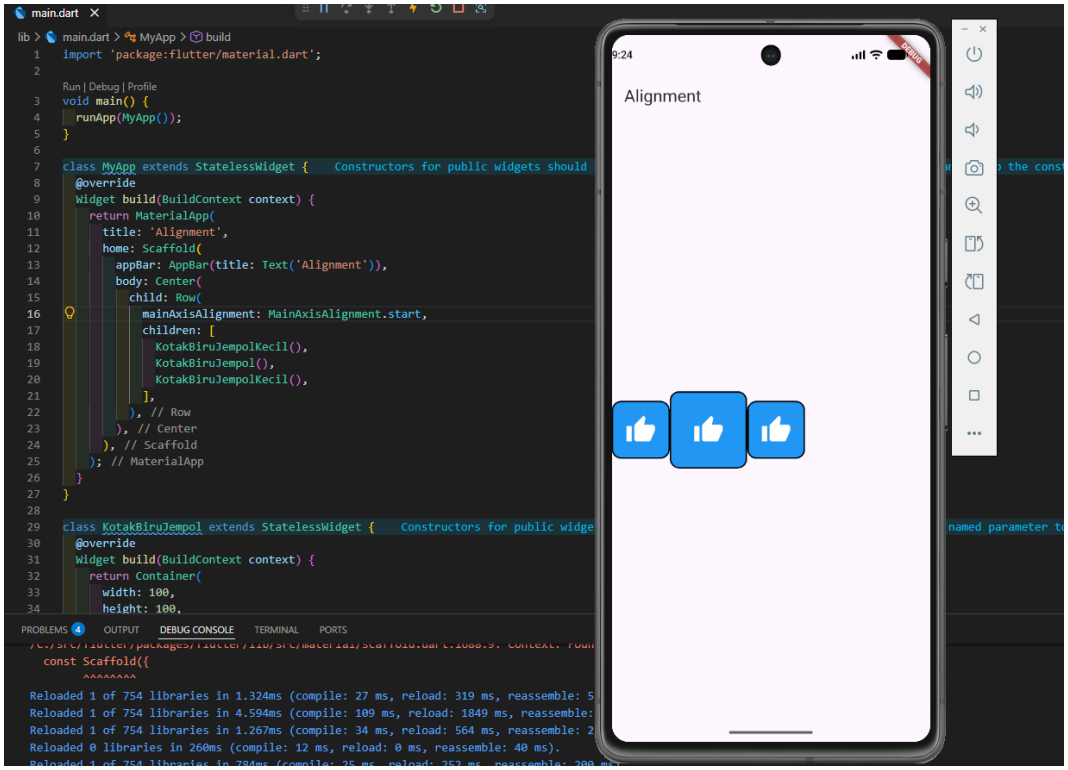

main.dart
lib > main.dart > MyApp > build
import 'package:flutter/material.dart';
Run | Debug | Profile
void main() {
runApp(MyApp());
}
class MyApp extends StatelessWidget {
Constructors for public widgets should
@override
Widget build(BuildContext context) {
return MaterialApp(
title: 'Alignment',
home: Scaffold(
appBar: AppBar(title: Text('Alignment')),
body: Center(
child: Column(
mainAxisSize: MainAxisSize
// Ganti ke MainAxisSize.min untuk melihat perubahannya
KotakBiruJempolKecil(),
SizedBox(height: 20), // jarak antar kotak (secara vertikal)
KotakBiruJempol(),
SizedBox(height: 20),
KotakBiruJempolKecil(),
],
), // Column
), // Center
), // Scaffold
); // MaterialApp
}
}
class KotakBiruJempol extends StatelessWidget {
Constructors for public widge
named par
@override
Widget build(BuildContext context) {
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
lib/main.dart:25:3: Error: Expected an identifier, but got ','.
Try inserting an identifier before ','.
),,
^
lib/main.dart:12:21: Error: Too many positional arguments: 0 allowed, but 1 found.
Try removing the extra positional arguments.
Alignment

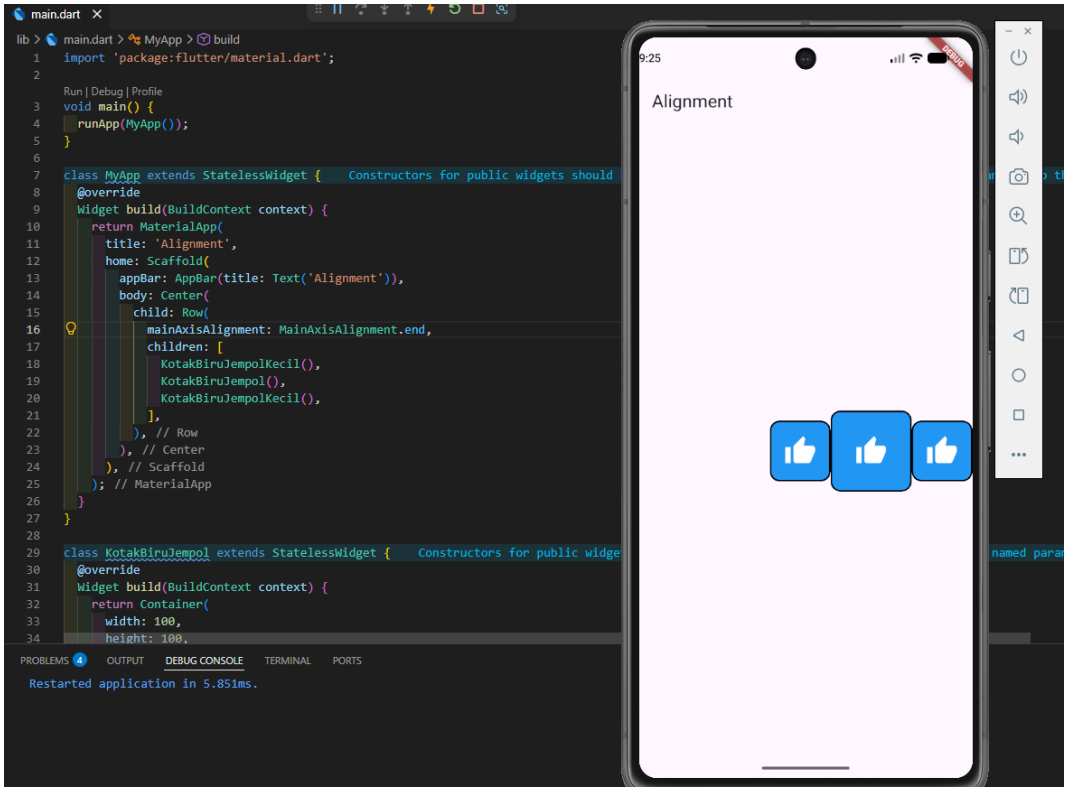


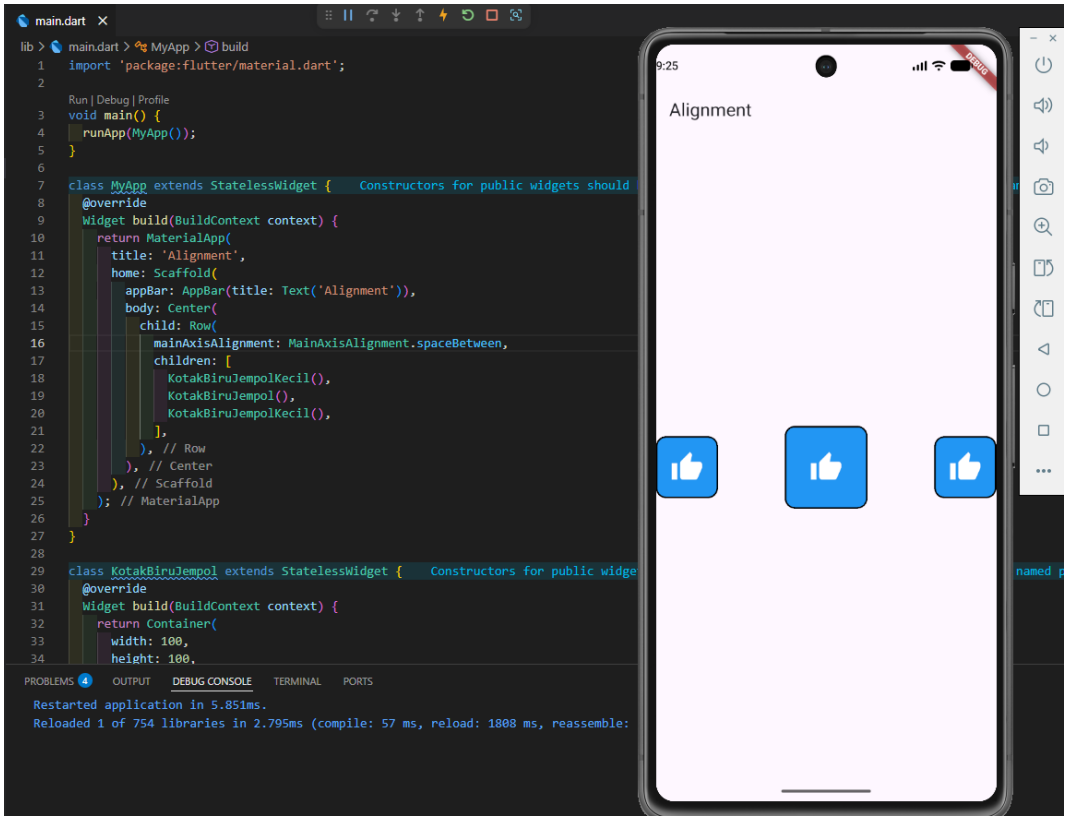
Keterangan Implementasi MainAxisSize pada Row dan Column untuk mengatur penggunaan ruang pada layout.

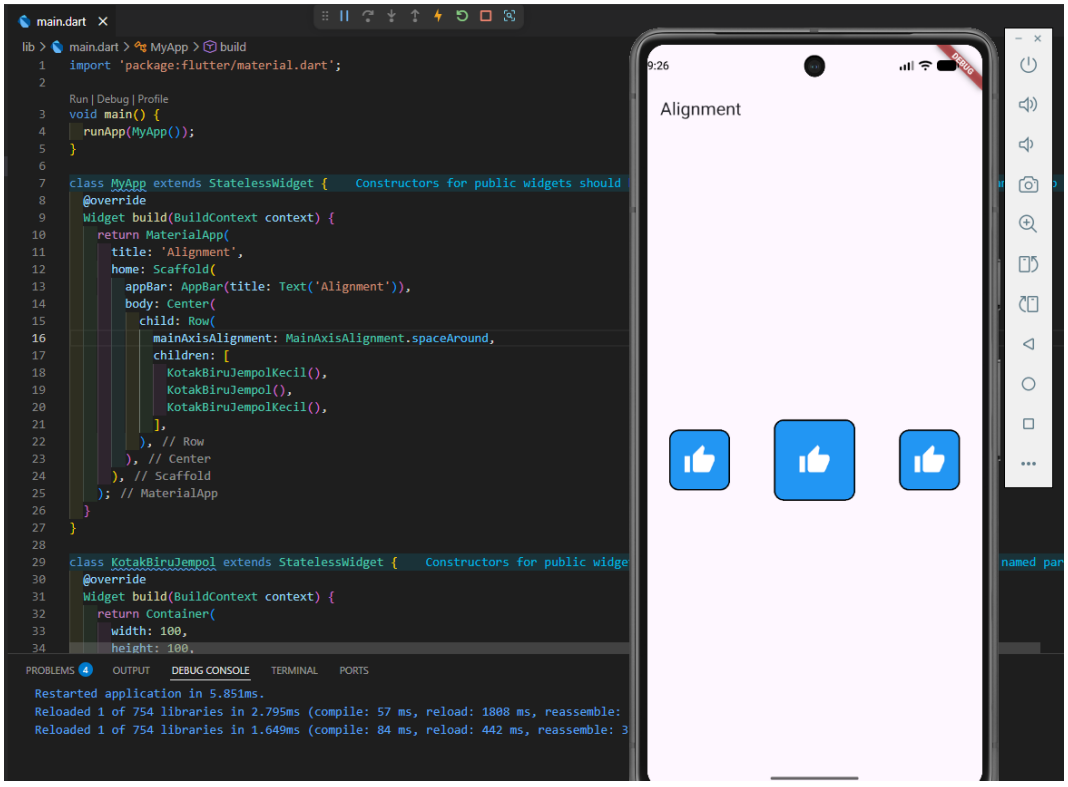
B. Percobaan MainAxisAlignment

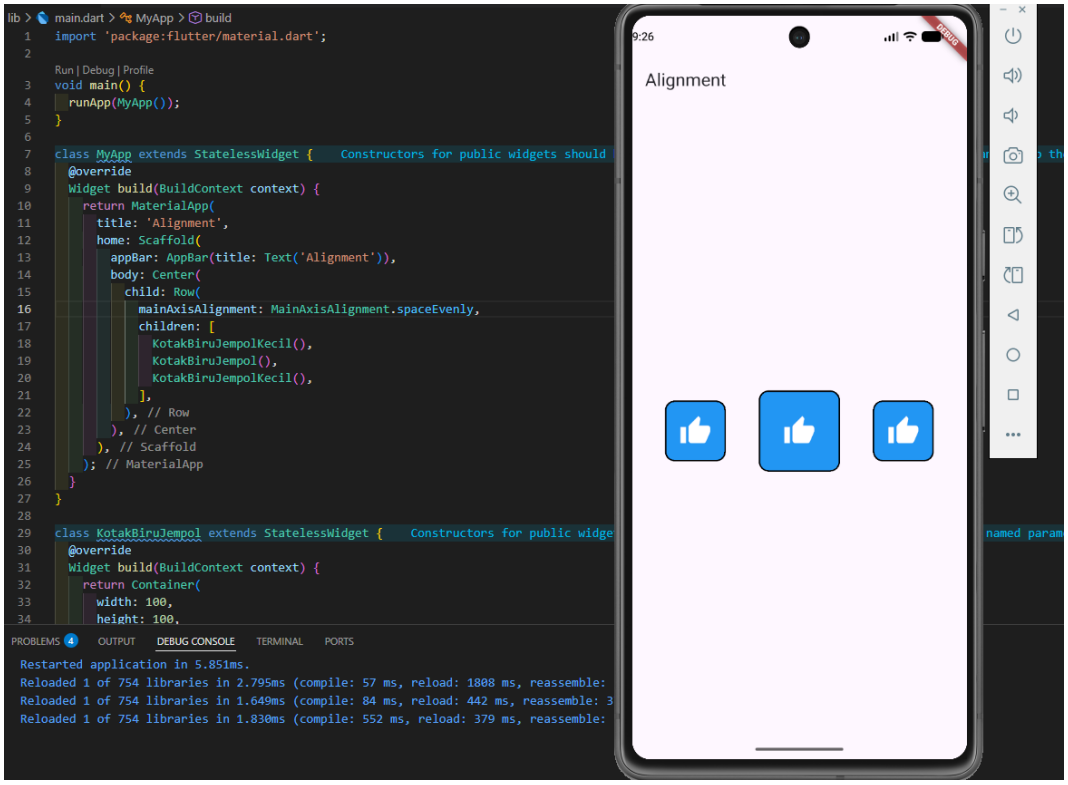



main.dart
lib > main.dart > MyApp > build
import 'package:flutter/material.dart';
Run | Debug | Profile
void main() {
runApp(MyApp());
}
class MyApp extends StatelessWidget {
@override
Widget build(BuildContext context) {
return MaterialApp(
title: 'Alignment',
home: Scaffold(
appBar: AppBar(title: Text('Alignment')),
body: Center(
child: Row(
mainAxisAlignment: MainAxisAlignment.start,
children: [
KotakBiruJempolKecil(),
KotakBiruJempol(),
KotakBiruJempolKecil(),
],
), // Row
), // Center
), // Scaffold
); // MaterialApp
}
}
class KotakBiruJempol extends StatelessWidget {
@override
Widget build(BuildContext context) {
return Container(
width: 100,
height: 100,
PROBLEMS 4
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
const Scaffold({
^^^^^^^^
Reloaded 1 of 754 libraries in 1.324ms (compile: 27 ms, reload: 319 ms, reassemble: 5
Reloaded 1 of 754 libraries in 4.594ms (compile: 109 ms, reload: 1849 ms, reassemble:
Reloaded 1 of 754 libraries in 1.267ms (compile: 34 ms, reload: 564 ms, reassemble: 2
Reloaded 0 libraries in 260ms (compile: 12 ms, reload: 0 ms, reassemble: 40 ms).
9:24
Alignment


main.dart
lib > main.dart > MyApp > build
import 'package:flutter/material.dart';
Run | Debug | Profile
void main() {
runApp(MyApp());
}
class MyApp extends StatelessWidget {
@override
Widget build(BuildContext context) {
return MaterialApp(
title: 'Alignment',
home: Scaffold(
appBar: AppBar(title: Text('Alignment')),
body: Center(
child: Row(
mainAxisAlignment: MainAxisAlignment.end,
children: [
KotakBiruJempolKecil(),
KotakBiruJempol(),
KotakBiruJempolKecil(),
],
), // Row
), // Center
), // Scaffold
); // MaterialApp
}
}
class KotakBiruJempol extends StatelessWidget {
@override
Widget build(BuildContext context) {
return Container(
width: 100,
height: 100,
PROBLEMS 4
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Restarted application in 5.851ms.
9:25
Alignment


main.dart
lib > main.dart > MyApp > build
1 import 'package:flutter/material.dart';
Run | Debug | Profile
3 void main() {
4 runApp(MyApp());
5 }
7 class MyApp extends StatelessWidget { Constructors for public widgets should
8 @override
9 Widget build(BuildContext context) {
10 return MaterialApp(
11 title: 'Alignment',
12 home: Scaffold(
13 appBar: AppBar(title: Text('Alignment')),
14 body: Center(
15 child: Row(
16 mainAxisAlignment: MainAxisAlignment.spaceBetween,
17 children: [
18 KotakBiruJempolKecil(),
19 KotakBiruJempol(),
20 KotakBiruJempolKecil(),
21],
22), // Row
23), // Center
24), // Scaffold
25); // MaterialApp
26 }
27 }
29 class KotakBiruJempol extends StatelessWidget { Constructors for public widge
30 @override
31 Widget build(BuildContext context) {
32 return Container(
33 width: 100,
34 height: 100,
PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Restarted application in 5.851ms.
Reloaded 1 of 754 libraries in 2.795ms (compile: 57 ms, reload: 1808 ms, reassemble:
9:25
Alignment
DEBUG


main.dart
lib > main.dart > MyApp > build
1 import 'package:flutter/material.dart';
Run | Debug | Profile
3 void main() {
4 runApp(MyApp());
5 }
7 class MyApp extends StatelessWidget { Constructors for public widgets should
8 @override
9 Widget build(BuildContext context) {
10 return MaterialApp(
11 title: 'Alignment',
12 home: Scaffold(
13 appBar: AppBar(title: Text('Alignment')),
14 body: Center(
15 child: Row(
16 mainAxisAlignment: MainAxisAlignment.spaceAround,
17 children: [
18 KotakBiruJempolKecil(),
19 KotakBiruJempol(),
20 KotakBiruJempolKecil(),
21],
22), // Row
23), // Center
24), // Scaffold
25); // MaterialApp
26 }
27 }
29 class KotakBiruJempol extends StatelessWidget { Constructors for public widge
30 @override
31 Widget build(BuildContext context) {
32 return Container(
33 width: 100,
34 height: 100,
PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Restarted application in 5.851ms.
Reloaded 1 of 754 libraries in 2.795ms (compile: 57 ms, reload: 1808 ms, reassemble:
Reloaded 1 of 754 libraries in 1.649ms (compile: 84 ms, reload: 442 ms, reassemble: 3
9:26
Alignment
DEBUG



Keterangan : Implementasi berbagai nilai MainAxisAlignment pada Row dan Column untuk mengatur posisi widget pada sumbu utama.

D. Tugas – Tampilan Informasi Cuaca

Nah, ini bagian yang **paling penting** karena merupakan hasil tugas Modul 3.

1. Kode Program

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Alignment',
      home: Scaffold(
        appBar: AppBar(title: Text('Alignment')),
        body: Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
```

```
            crossAxisAlignment: CrossAxisAlignment.center,
            children: [
              const Text(
                'Malang',
                style: TextStyle(fontSize: 40),
              ),
              const SizedBox(height: 30),
              const Text(
                '25°',
                style: TextStyle(fontSize: 75, fontWeight: FontWeight.bold),
              ),
              const SizedBox(height: 50),
              Row(
                mainAxisAlignment: MainAxisAlignment.spaceEvenly,
                crossAxisAlignment: CrossAxisAlignment.center,
                children: [
                  Column(
                    crossAxisAlignment: CrossAxisAlignment.center,
                    children: const [
                      Text('Minggu', style: TextStyle(fontSize: 16)),
                      SizedBox(height: 10),
                      Icon(Icons.sunny, size: 40),
                      SizedBox(height: 10),
                      Text('20°C', style: TextStyle(fontSize: 16)),
                    ],
                  ),
                  Column(
                    crossAxisAlignment: CrossAxisAlignment.center,
                    children: const [
                      Text('Senin', style: TextStyle(fontSize: 16)),
                      SizedBox(height: 10),
                      Icon(Icons.cloudy_snowing, size: 40),
                      SizedBox(height: 10),
                      Text('23°C', style: TextStyle(fontSize: 16)),
                    ],
                  ),
                  Column(
                    crossAxisAlignment: CrossAxisAlignment.center,
                    children: const [
                      Text('Selasa', style: TextStyle(fontSize: 16)),
                      SizedBox(height: 10),
                      Icon(Icons.cloud, size: 40),
                      SizedBox(height: 10),
                      Text('22°C', style: TextStyle(fontSize: 16)),
                    ],
                  ),
                ],
              ),
            ],
          ),
        ),
```

```
        ),
      );
    }
}
```

Keterangan Implementasi antarmuka informasi cuaca menggunakan kombinasi Column dan Row serta pengaturan MainAxisAlignment.spaceEvenly.

2. Hasil Tampilan



Keterangan : Hasil implementasi antarmuka informasi cuaca dengan informasi kota, suhu, dan prakiraan cuaca menggunakan beberapa icon cuaca.

IV. Kesimpulan

- Properti mainAxisSize mengendalikan konsumsi luas ruang oleh Row atau Column pada sumbu utamanya, apakah ingin memenuhi layar (*max*) atau hanya membungkus seukuran kontennya (*min*).
- Properti mainAxisAlignment mengendalikan distribusi, perataan letak, dan jarak spasi *widget* anak di sepanjang sumbu utama.
- Properti crossAxisAlignment mengendalikan penempatan posisi *widget* pada sumbu tegak lurus (*cross axis*) terhadap sumbu utama.

- Kombinasi strategis antara Row, Column, dan pengaturan properti *alignment* tersebut sangat vital dalam menyusun desain antarmuka (*layout*) yang rapi, fleksibel, dan responsif terhadap dimensi layar perangkat.